

Worksheet: Measure heat capacity with a calorimeter

Name(s) _____

In this experiment, a hot metal cube is dropped into an insulated container of water. The increase in the water temperature is used to determine the heat capacity of the metal.

Student Learning Objectives

1. Explain the concept of heat capacity.
2. Apply the concept of an adiabatic process.
3. Apply the first law for a closed system.
4. Be aware of assumptions made and sources of error in using a static calorimeter.
5. Learn proper use and calibration of calorimeters.
6. Apply statistics to determine heat capacity from repeat experiments and determine standard deviation.

Equipment

A well-insulated calorimeter that contains 1000 mL of water, a stirrer, and a thermometer to measure the water temperature.

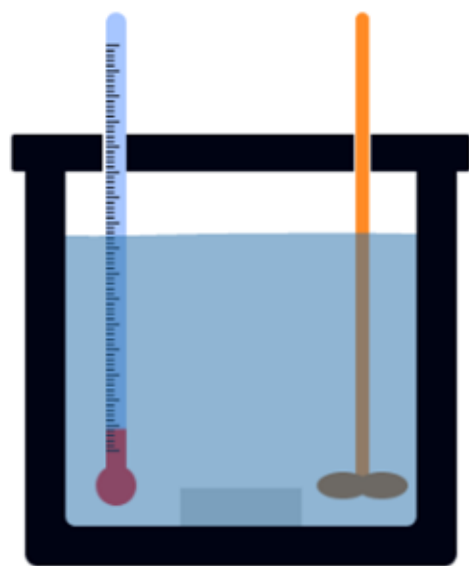
initial water temperature (°C): 4.0

initial block temperature (°C): 30

mass of block (g): 400

metal:

heat capacity: 0.133 J/(g K)



Assumptions

1. The combined metal-water-calorimeter system is adiabatic during the experiment.
2. The water is well mixed, so the measured water temperature is uniform.
3. The heat capacity of the metal sample is independent of temperature.

Notation

m_{Pt} = mass of Pt added to calorimeter (g)

m_W = mass of water in calorimeter (1000 g)

m_{sample} = mass of sample (g)

C_{Pt} = heat capacity of Pt = 0.133 J/(g °C)

C_W = water heat capacity = 4.184 J/(g °C)

C_{cal} = effective heat capacity of calorimeter (J/°C)

C_{sample} = heat capacity of sample (J/(g °C))

T_{sample} = starting temperature of sample (°C)

T_W = initial temperature of water in calorimeter (°C)

T_{final} = final equilibrium temperature (°C)

Questions to answer before starting experiment

1. In this experiment, 1000 mL of water is in an insulated container. What happens to the final temperature change of the water if more water is used? How does this affect the accuracy of the heat capacity measurement?
2. What happens to the water temperature change if the mass of metal is increased? What practical problem occurs if too much metal is used?
3. How does increasing the initial metal temperature affect the experiment? What assumptions may become less valid at high temperature?

Calibrate the calorimeter to determine the effective heat capacity of the calorimeter.

The heat absorbed by the container, thermometer, and stirrer is $C_{cal}(T_{final} - T_W)$ since these parts of the calorimeter start at the same temperature as the water.

1. Start the stirrer.
2. Use the slider to select the initial water temperature T_W and record the value in Table 1.
3. Select the metal Pt, select its initial temperature and its mass using the sliders, and record their values in Table 1.
4. Drop the metal into the calorimeter. Record the water temperature every 10 s until it changes by less than 0.1 °C. Use this as the equilibrium temperature T_{final} and record the value in Table 1.
5. Calculate the heat gained by the calorimeter to determine C_{cal} .

Energy balance: $Q_{lost\ by\ metal} = Q_{gained\ by\ water} + Q_{gained\ by\ calorimeter}$

$$m_{Pt}C_{Pt}(T_{Pt} - T_{final}) = m_W C_{PW}(T_{final} - T_W) + C_{cal}(T_{final} - T_W)$$

$$C_{Pt} = 0.133\text{ J/(g °C)} \quad C_{PW} = 4.184\text{ J/(g °C)}$$

Table 1 Effective heat capacity of calorimeter

T_W (°C)	m_W (g)	T_{Pt} (°C)	m_{Pt} (g)	T_{final} (°C)	C_{cal} (J/°C)
	1000				

Measure heat capacity of unknown sample

1. Click reset.
2. Select a sample (A, B, C, D, E) from the drop-down menu and record the letter in Table 2.
3. Turn on stirrer.
4. Select the initial water temperature T_W and record in Table 2.
5. Select the initial temperature of the metal block T_{sample} and record it in Table 2.
6. Select the mass of the metal block m_{sample} and record the value in Table 2.
7. Drop the metal into the water. Record the water temperature every 10 s until it changes by less than 0.1 °C. Use this as the equilibrium temperature T_{final} and record the value in Table 2.
8. Repeat the experiment 3 more times for the same sample but for different starting values of T_W , T_{sample} , and m_{sample} , and record the values and results in Table 2.
10. Use the energy balance to calculate the heat capacity $C_{Psample}$ of the sample:

$$m_{sample}C_{Psample}(T_{sample} - T_{final}) = m_W C_{PW}(T_{final} - T_W) + C_{cal}(T_{final} - T_W)$$

and record the heat capacity in Table 2.

Table 2 Heat capacity measurements for sample _____

Exp.	T_W (°C)	T_{sample} (°C)	m_{sample} (g)	T_{final} (°C)	$C_{Psample}$ J/(g °C)
1					
2					
3					
4					

11. Calculate the average value of heat capacity and standard deviation.

$$C_{Psample} = \text{_____} \pm \text{_____}$$

12. Compare your heat capacity to known values. Which metal is most likely your sample?

Table 3 Heat capacities of metals

Metal	$C_{Psample}$ (J/(g K))
Au	0.13
Pb	0.13
Ag	0.23
Cu	0.39
Fe	0.45
Al	0.90

Questions to answer

1. In the energy balance equation, which temperature differences are positive?

$$(T_{sample} - T_{final})$$

$$(T_{final} - T_W)$$

2. Does the stirrer add energy to the water? If so, should this energy be included in the first-law balance? Under what conditions can it be neglected?

3. If heat is lost to the surroundings, will the calculated heat capacity of the metal be overestimated or underestimated?
4. How could you experimentally test the assumption that the system is adiabatic? What data would you need?
5. For each error source listed below, state whether the calculated heat capacity would be too high, too low, or uncertain:
- heat loss _____
- thermometer lag _____
- incomplete mixing _____
- wrong metal mass _____
- wrong water mass _____
- heat absorbed by the calorimeter _____